

METHODS

Accelerated Method for Determining Ischemic Brain Damage in Rats and Assessment the Neuroprotective Effect of a Complex Herbal Remedy

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The severity of ischemic injury was evaluated by densitometry of brain samples stained with 2,3,5-triphenyltetrazolium chloride (TTC) on a rat model of cerebral ischemia/reperfusion (common carotid artery occlusion) and the neuroprotective activity of an extract of *Astragalus membranaceus*, *Scutellaria baicalensis*, and *Phlojodicarpus sibiricus* was assessed. Occlusion of the common carotid arteries led to a weakening of TTC staining of the brain tissue: densitometric indicators of the staining intensity for the cortex and striatum were lower than the corresponding indicators of sham-operated rats by 18.3 and 10.4%. The mean intensity of staining of brain samples did not differ in rats treated with the extract and sham-operated animals, which attested to its neuroprotective effect. The applied method is convenient for evaluation of the severity of ischemic brain damage at the early stages and screening potential neuroprotective agents.

Key Words: cerebral ischemia/reperfusion; densitometry; herbal remedy; neuroprotection

Different models of cerebral ischemia are used in experimental medicine. The most often used models are middle cerebral artery occlusion in rats and standard pathomorphological techniques. These experiments are time consuming and require special skills and microscopic and pathological techniques. Simple and at the same time informative methods are needed for rapid evaluation of the severity of ischemic brain damage and screening of neuroprotective agents.

The purpose of this work was to determine the degree of ischemic brain damage at the early stages by staining brain tissue samples with 2,3,5-triphenyltetrazolium chloride (TTC) followed by densito-

metric analysis, and to evaluate the neuroprotective effect of a standardized extract from *Astragalus membranaceus*, *Scutellaria baicalensis*, and *Phlojodicarpus sibiricus* [1].

MATERIALS AND METHODS

The experiments were carried out on male and female Wistar rats ($n=15$) weighing 200-220 g. The study was performed in compliance with the rules for handling laboratory animals [2] and was approved by the Local Ethics Committee of Institute of General and Experimental Biology (Protocol No. 3 of June 10, 2023).

The animals were divided into three groups (5 animals in each): modeling of ischemia/reperfusion (IR; group 1), rats received the extract before IR modeling (IR+extract; group 2), and sham-operated animals (SO; group 3). Group 2 animals received the extract in a

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dose of 200 mg/kg for 7 days before surgery. Rats of groups 1 and 3 received purified water according to the same scheme. IR was modeled by applying micro-vascular clips simultaneously to both common carotid arteries in rats anesthetized with sodium thiopental (45 mg/kg). The clips were removed after 10 min and blood flow resumption was controlled. The surgical wound was sutured and treated with an antiseptic, and the rats were placed in home cages. Group 3 rats (SO) were subjected to the same surgical manipulations except clip application.

The animals were decapitated 24 h after surgery, and the brain was isolated on ice. Coronal brain sections (2 mm thick) at the level of +0.48 mm from bregma (Paxinos and Watson, 1982) were made, placed in 2% TTC (Sigma) for 30 min at 37°C, and fixed in a 10% solution of neutral formaldehyde (pH 7.4) [3]. Images of brain samples were obtained using a Canon EOS 600D digital camera. The staining of brain areas (cortex and striatum) was evaluated by densitometry of scanned images converted to an 8-bit grayscale (Epson M200 scanner) (Fig. 1) [4].

Statistical processing of the obtained results was carried out by using the Statistica 8.0 software (StatSoft, Inc.). The significance of differences was deter-

TABLE 1. Effect of the Herbal Extract on the Intensity of Staining of the Cortex and Striatum of Brain Coronal Sections of Rats with IR (% on the scale; $M \pm m$)

Group	Cortex	Striatum
IR	63.7 $\pm$ 1.5 ⁺	68.2 $\pm$ 4.2 ⁺
Extract+IR	77.1 $\pm$ 1.6 [*]	76.5 $\pm$ 1.0 [*]
SO	78.0 $\pm$ 3.2	76.1 $\pm$ 2.6

Note. $p < 0.05$ in comparison with *IR, ⁺SO.

mined using Student's *t* test. The data were presented as $M \pm m$. The differences were considered statistically significant at $p < 0.05$.

RESULTS

The brain tissue of SO and experimental rats was uniformly stained dark-red with TTC (Fig. 1, *a*). Visual assessment revealed less intense staining of the brain tissue in group 1 rats (IR).

According to densitometry data, IR of the brain led to a decrease in the intensity of TTC staining of the cortex and striatum by 18.3 and 10.4% in comparison with the corresponding indicators in SO animals

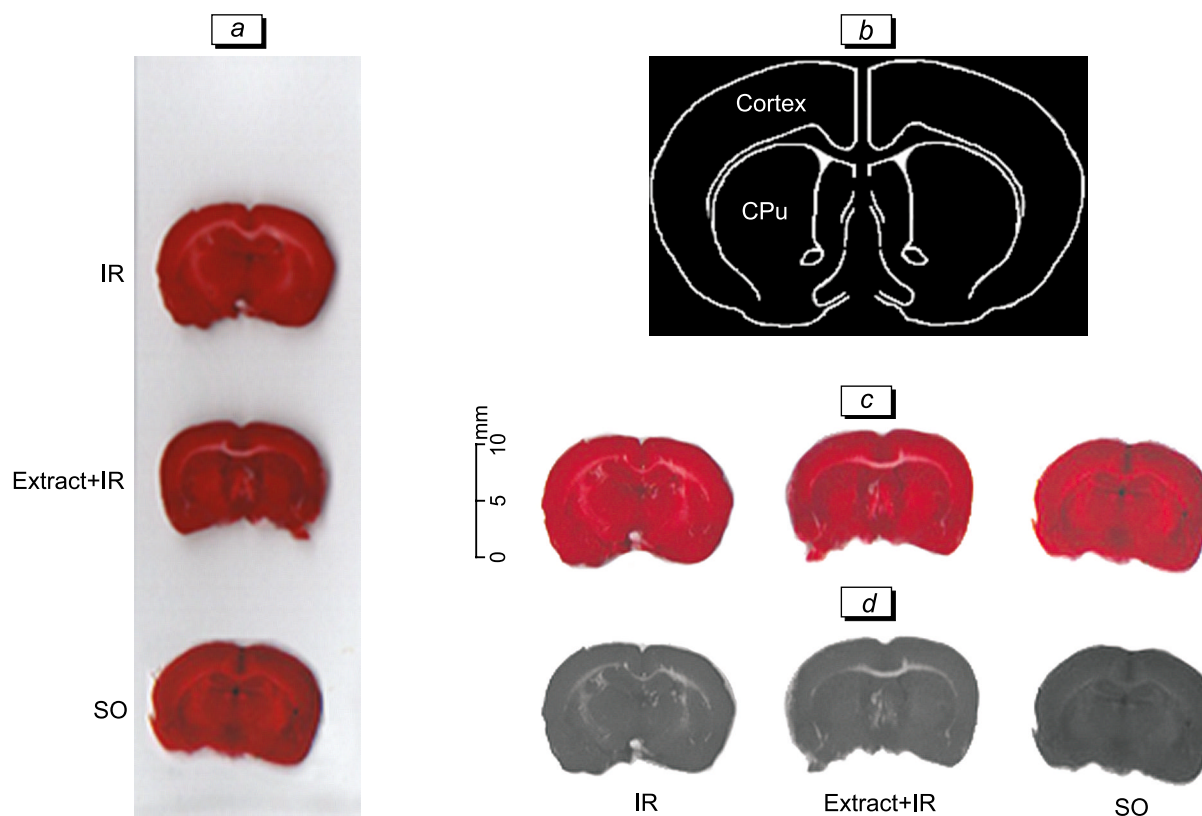


Fig. 1. Representative coronal sections of rat brain stained with TTC. *a*) Photographs of coronal sections of the brain; *b*) diagram of a coronal section of the rat brain; CPu (caudate putamen): striatum (Paxinos and Watson, 1982); *c*) 24-bit color scanned sections: intense dark red areas correspond to healthy tissue; *d*) the same images converted to an 8-bit grayscale to quantify the staining intensity.

(Table 1). In rats receiving the extract, these values did not differ from those in the SO group.

The method of cerebral tissue staining with TTC dye is widely used to determine the volume of focal ischemic damage of the brain [5]. TTC is reduced by respiratory enzymes in healthy tissues to insoluble formazan that stain viable areas dark red. In this case, TTC molecules serve as proton acceptors for dehydrogenases, which together with cytochromes are components of the respiratory chain of electron transfer of the inner mitochondrial membrane [3]. Accordingly, the ischemic or necrotic area in the brain does not stain or stains weakly with the dye due to decreased enzyme activity, indicating initial ischemic damage to neurons or their death.

The degree of damage in cerebral IR was assessed by the mean intensity of TTC staining using densitometry of scanned images of samples. Coronal cerebral sections of rats subjected to IR were stained significantly weaker than samples of SO rats due to a decrease in dehydrogenase activity and diffuse metabolic changes in the brain tissue, mainly in the cortex and striatum, which served as a diagnostic indicator of initial ischemic disorders. The initial decrease of staining intensity is due to intensive metabolism of neurons in these structures and high sensitivity of these areas to ischemia. Course administration of the extract to rats prevented the development of metabolic changes in the nervous tissue at bilateral occlusion of the common carotid arteries. Pharmacological correction of mitochondrial functions can increase the energy potential and improve neuronal survival at the early stages of cerebral ischemia and serve as an integral indicator of the anti-ischemic effect of the drug. The neuroprotective effect of the standardized extract is determined by the presented a complex of active substances: flavonoids, astragalosides, and coumarins [1,6,7].

Thus, densitometry of TTC stained brain samples makes it possible to relatively quickly determine the degree of ischemic brain damage at the early stages

in bilateral occlusion of the common carotid arteries and to evaluate the neuroprotective effect of the test drug immediately after the end of the experiment using simple techniques, a desktop scanner, and without routine histological studies.

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Conflict of interest. The authors have no conflicts of interest to declare.

REFERENCES

1. Nikolaev SM, Nikolaeva GG, Nikolaeva IG, Radnaeva LD, Turtueva TA, Taraskin VV, Guljaev SM, Urbanova EZ. Patent RU No. 2603465. Method for producing agent possessing anti-ischemic and antioxidant activity. Bull. No. 33. Published November 27, 2016.
2. Manual for Preclinical Studies of New Pharmacological Substances. Part I, Mironov AN, ed. Moscow, 2012. Russian.
3. Bederson JB, Pitts LH, Tsuji M, Nishimura MC, Davis RL, Bartkowski H. Rat middle cerebral artery occlusion: evaluation of the model and development of a neurologic examination. *Stroke*. 1986;17(3):472-476. doi: 10.1161/01.str.17.3.472
4. Mathews KS, McLaughlin DP, Ziabari LH, Toner CC, Street PC, Hisgrove E, Bezzina EL, Stamford JA. Rapid quantification of ischaemic injury and cerebroprotection in brain slices using densitometric assessment of 2,3,5-triphenyltetrazolium chloride staining. *J. Neurosci. Methods*. 2000;102(1):43-51. doi: 10.1016/S0165-0270(00)00277-6
5. Popp A, Jaenisch N, Witte OW, Frahm C. Identification of ischemic regions in a rat model of stroke. *PLoS One*. 2009;4(3):e4764. doi: 10.1371/journal.pone.0004764
6. Gulyaev SM, Zhalsanov YuV, Turtueva TA, Nikolaeva GG. Antihypoxic activity and anti-ischemic effect of complex herbal remedy. *Obzory Klin. Farmakol. Lek. Ter*. 2022;20(3):325-330. Russian. doi: 10.17816/RCF203325-330
7. Gulyaev SM, Fedorova AV, Lemza SV, Mondodoev AG, Urbanova EZ, Nikolaev SM. Influence of the *Phlojodicarpus sibiricus* extract on energetic status of rat brain cells after carotid artery occlusion. *Sib. Med. Zh. (Irkutsk)*. 2015;139(8):60-63. Russian.